

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 01.1

Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

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Aim

To determine the minimum filter order and cutoff frequency and analyze the magnitude, phase, impulse response, step response, group delay, and pole-zero characteristics of a Butterworth IIR low-pass filter.

Procedure / Calculations

Define the sampling and edge frequencies, normalize the frequencies, determine the minimum order with `buttord()`, design the filter with `butter()`, and analyze the frequency, time-domain, group-delay, and pole-zero responses.

MATLAB Program

```
clc; clear

// NO:1 Butterworth IIR Filters
// TE: 1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);

freqz(b,a)           % Magnitude & Phase

figure
[h,n]=impz(b,a,50);
stem(n,h); grid on;
title('Impulse Response')

figure
stepz(b,a)

title('Step Response')

figure
zplane(b,a)

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);

freqz(b,a)          % Magnitude & Phase

figure
[h,n]=impz(b,a,50);
...title('Impulse Response')

figure
stepz(b,a)

figure
zplane(b,a)

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn); Input:
```

- Passband Ripple (R_p) = 1 dB
- Stopband Attenuation (R_s) = 20 dB
- Passband Frequency (F_p) = 1000 Hz
- Stopband Frequency (F_s) = 3000 Hz
- Sampling Frequency ($F_{sampling}$) = 8000 Hz
- Filter Choice = Low-Pass Filter (LPF)

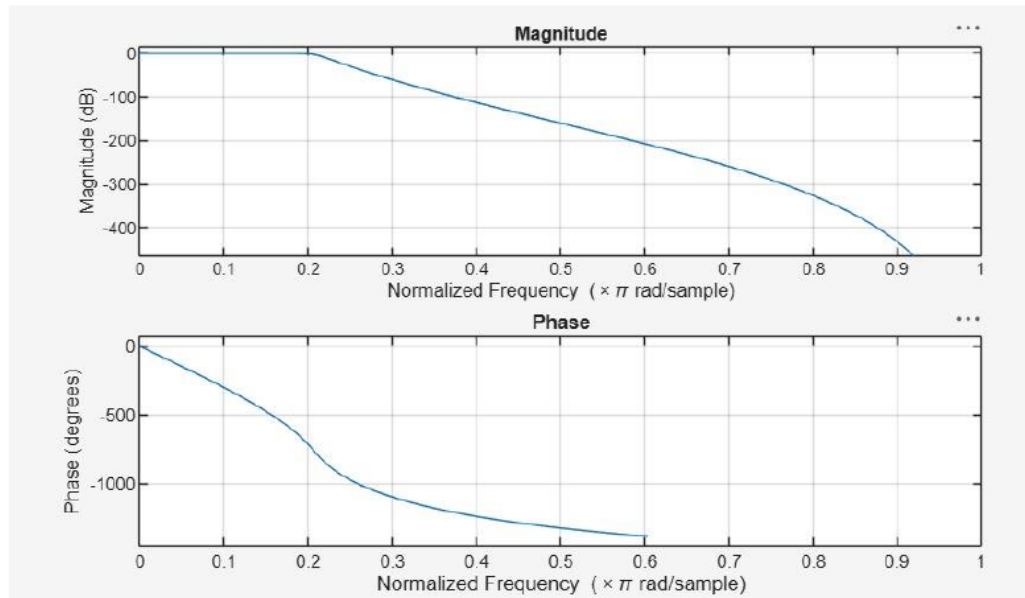
Output:

Objective 1:

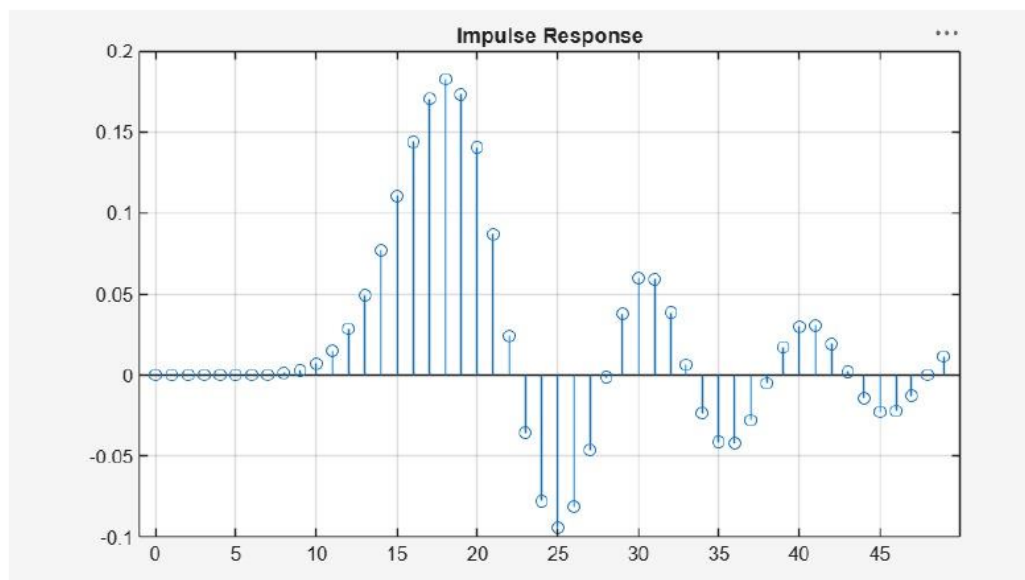
Minimum Filter Order (N) = 17

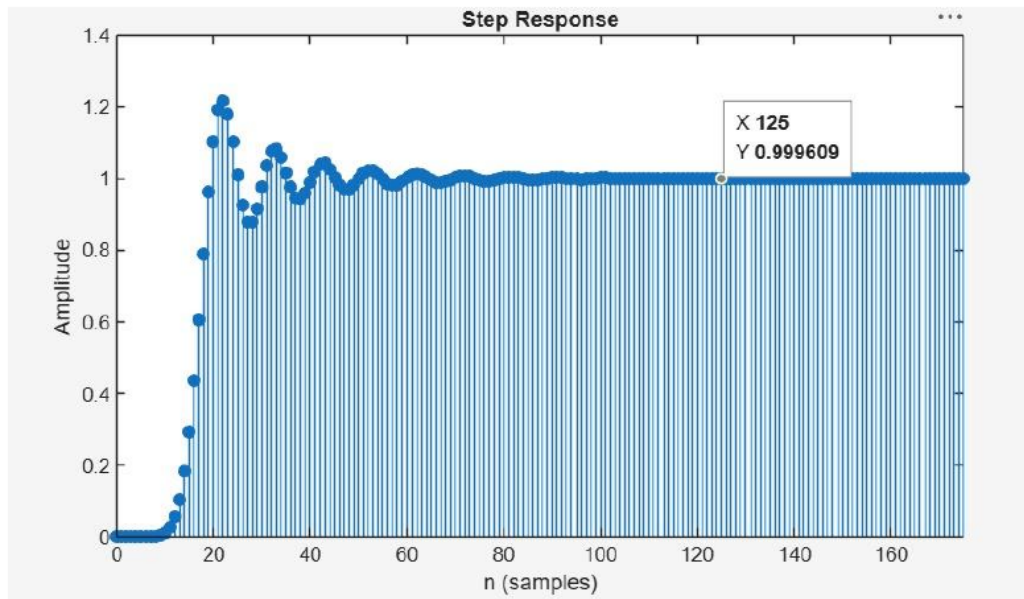
Cutoff Frequency (Wn) = 0.2083

Objective 2:

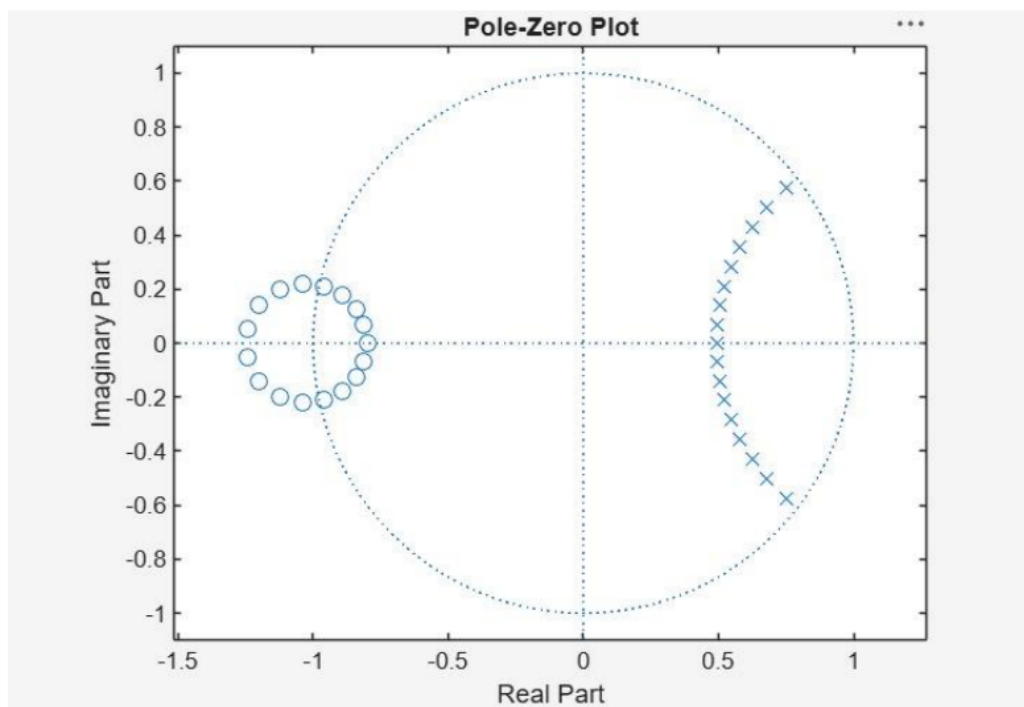


Objective 3:





Objective 4:



Result:

- 1.The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.
- 2.The magnitude and phase responses of the Butterworth IIR low-pass filter were obtained and analyzed.
- 3.The impulse response, step response, and group delay characteristics were evaluated successfully.
4. The pole-zero plot confirmed the stability and proper performance of the designed filter.

Practical Question – 1

Design a **digital Butterworth IIR Low-Pass Filter** using MATLAB to meet the following specifications: Passband ripple of **1 dB**, minimum stopband attenuation of **40 dB**, passband edge frequency of **1000 Hz**, stopband edge frequency of **1500 Hz**, and sampling frequency of **8000 Hz**. Determine the minimum filter order and cutoff frequency required to satisfy the given specifications. Obtain and analyze the magnitude and phase responses of the designed filter. Also, generate the pole-zero plot and verify the stability and performance characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question – 2

Design a **digital Butterworth IIR Low-Pass Filter** using MATLAB for a signal processing application with a passband ripple of **0.5 dB**, stopband attenuation of **50 dB**, passband edge frequency of **1200 Hz**, stopband edge frequency of **1800 Hz**, and sampling frequency of **10000 Hz**. Calculate the minimum filter order and corresponding cutoff frequency required to satisfy the design requirements. Plot the magnitude and phase responses of the filter and evaluate its frequency-selective characteristics. Generate the pole-zero plot and comment on the stability and effectiveness of the designed low-pass filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

The minimum filter order and cutoff frequency were determined successfully. The magnitude, phase, impulse and step responses were obtained, and the pole-zero plot verified stable filter behavior.

Student Details

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Course: ECA09 - Digital Signal Processing